

**Discussion on initial reputation calculation
using Bayesian modelling & intuition behind
threat score threshold values**

DRONE REPUTATION DYNAMICS

Intuition of Reputation (R)

Reputation is a metric ranging from 0 to 1 that quantifies source reliability.

- **R = 1:** Fully Trusted. A highly reputable source with a clean history.
- **R = 0:** Fully Untrusted. A malicious source identified by the system.

Feedback Mechanism

Reward: If transmitted data is *non-malicious*, R increases.

Penalty: If data is found to be *infected*, R decreases significantly.



| The "Cold Start" Challenge



Unknown Origin

When a drone enters the network for the **first time**, there is no historical data in the database to compute its reputation.



Security Risk

Assigning a static default reputation is dangerous. A malicious drone could exploit a high default score to bypass deep inspection.



Contextual Strategy

We solve this by using **Observable Contextual Features** (Zone & File Type) to derive a principled prior reputation.

BAYESIAN CONTEXTUAL STRATEGY

Observable Features

Even for new drones, two critical attributes are always known at the ingestion point:

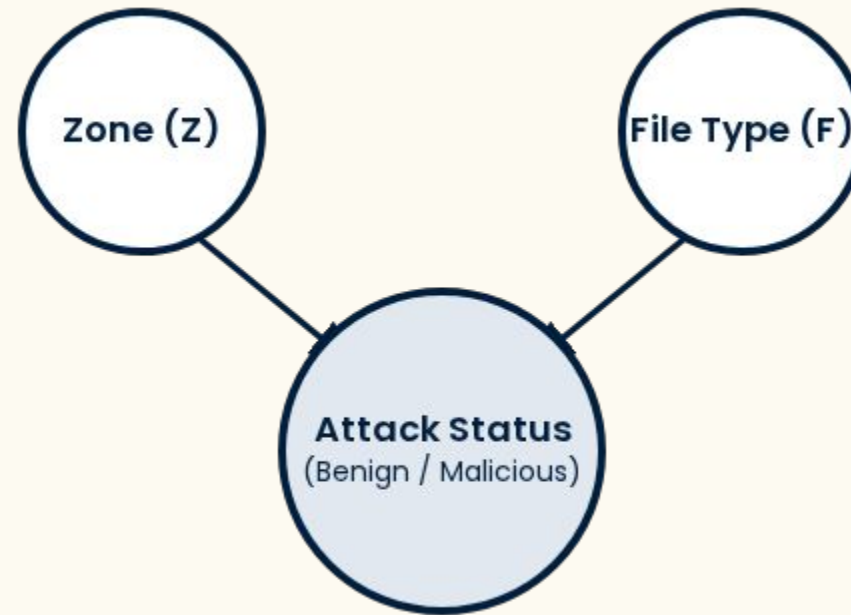
- **Zone (Z):** The operational zone from which data originates.
- **File Type (F):** The type of data (Video, ZIP, Image, etc.).

Contextual Prior

The model answers: *"Given a file is from Zone Z and has File Type F, what is the probability it is benign?"*

This provides a data-driven reputation for new drones without relying on history.

BAYESIAN BELIEF NETWORK (BBN)



Topology: Observable evidence (Zone & File Type) influences the probability of Attack Status.

BAYESIAN MATHEMATICAL FORMULATION

We compute the posterior probability $P(N \mid \text{Zone}, \text{File})$ to derive the reputation score R .

$$R = \frac{P(N) \cdot P(\text{Zone} \mid N) \cdot P(\text{File} \mid N)}{[P(N) \cdot P(\text{Zone} \mid N) \cdot P(\text{File} \mid N)] + [P(A) \cdot P(\text{Zone} \mid A) \cdot P(\text{File} \mid A)]}$$

N: No Attack (Benign) | A: Attack (Malicious)

Conditional Probability Table (CPT)

Table 1: Prior Distribution

Attack Status	Probability
N (No Attack)	$P(N)$
A (Attack)	$P(A) = 1 - P(N)$

Table 2: Zone Likelihood

Zone	$P(Z N)$	$P(Z A)$
Zone 1	$P(Z=1 N)$	$P(Z=1 A)$
Zone 2	$P(Z=2 N)$	$P(Z=2 A)$
...	...	...
Zone n	$P(Z=n N)$	$P(Z=n A)$

Table 3: File Type Likelihood

File Type	$P(F N)$	$P(F A)$
Video	$P(F=\text{video} N)$	$P(F=\text{video} A)$
Image	$P(F=\text{image} N)$	$P(F=\text{image} A)$
ZIP	$P(F=\text{zip} N)$	$P(F=\text{zip} A)$

EXAMPLE:

Metric	Benign (N) Probability	Attack (A) Probability
Prior Probability	$P(N) = 0.85$	$P(A) = 0.15$
Zone 3 Likelihood	$P(\text{Zone}=3 \mid N) = 0.12$	$P(\text{Zone}=3 \mid A) = 0.35$
ZIP File Likelihood	$P(\text{File}=\text{ZIP} \mid N) = 0.08$	$P(\text{File}=\text{ZIP} \mid A) = 0.45$

$$R = \frac{P(N) \cdot P(\text{Zone} \mid N) \cdot P(\text{File} \mid N)}{[P(N) \cdot P(\text{Zone} \mid N) \cdot P(\text{File} \mid N)] + [P(A) \cdot P(\text{Zone} \mid A) \cdot P(\text{File} \mid A)]}$$

Numerator: $0.85 \times 0.12 \times 0.08 = 0.00816$

Denominator: $0.00816 + (0.15 \times 0.35 \times 0.45) = 0.03179$

Final Reputation $R = 0.257$ (25.7%)

ADAPTIVE T_S THRESHOLDS

Initial Fixed Strategy Boundaries

- $T_S < 0.4 \rightarrow$ **Signature**
- $0.4 \leq T_S < 0.7 \rightarrow$ **ML Classifier**
- $T_S \geq 0.7 \rightarrow$ **Sandbox**

⚠ Static thresholds may fail under evolving attack patterns.

Solution

Use the **Security Feedback Loop** to adapt thresholds dynamically.

🔄 Continuous Optimization

Thresholds adjust based on system performance.

SYSTEM PERFORMANCE MONITORING

Detection Metrics

- False Positive Rate (FPR)
- False Negative Rate (FNR)
- Precision & Recall

Resource Metrics

- Sandbox Utilization
- Latency
- Throughput

Target Objectives

- FPR < 5%
- FNR < 3%
- Sandbox Load < 70%

(These are just examples)

SOFT THRESHOLD UPDATE RULE

Use a small learning rate (η) for gradual updates to θ_{low} and θ_{high} .

$$\theta_{\text{new}} = \theta_{\text{old}} + \eta \times (\text{Performance Error})$$

Example Update for θ_{high}

$$\theta_{\text{high,new}} = \theta_{\text{high,old}} + \eta(\text{FPR} - \text{FPR}_{\text{target}}) - \eta(\text{FNR} - \text{FNR}_{\text{target}})$$

High FPR: Increase thresholds → reduce false alarms.

High FNR: Decrease thresholds → improve detection.

FINAL ADAPTIVE WORKFLOW

- 1 Monitor detection outcomes
- 2 Compute deviation from targets
- 3 Apply soft update using η
- 4 Clamp thresholds within safe bounds

Why Soft Updates?

- Prevents sandbox overload
- Ensures stability
- Gradual convergence
- No oscillatory behaviour

